LAB_01

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2026-08-14

Laboratório 01 — Importação e Manipulação de Dados em R

Exercicio 1

Carregamento do pacote `tidyverse`.

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Exercicio 2

```r
View(table1)   #Se encontra no formato Tidy, pois as observacoes sao linhas
                #e as variaveis sao colunas.
```

```r
View(table2)  #Nao se encontra no formato Tidy, pois as variaveis cases
                #e population estao englobadas na variavel type.
```

```r
View(table3) #Nao se encontra no formato Tidy, pois a variavel rate engloba
              #cases e population.
```

```r
View(table4a) #Nao se encontra no formato Tidy, pois o tibble em questao possui
              #observacoes como variaveis.
```

```r
View(table4b) #Nao se encontra no formato Tidy, pois o tibble em questao possui
              #observacoes como variaveis.
```

Exercicio 3

```
taxas <- table1 |>
  mutate(rate = cases / population * 10000)




taxas
```

```
## # A tibble: 6 x 5
##   country      year  cases population  rate
##   <chr>       <dbl>  <dbl>      <dbl> <dbl>
## 1 Afghanistan  1999    745   19987071 0.373
## 2 Afghanistan  2000   2666   20595360 1.29
## 3 Brazil       1999  37737  172006362 2.19
## 4 Brazil       2000  80488  174504898 4.61
## 5 China        1999 212258 1272915272 1.67
## 6 China        2000 213766 1280428583 1.67
```

Exercicio 4

```
casos_por_year <- table1 |>
  group_by(year) |>
  summarise(cases = sum(cases))
```

Exercicio 5

```
casos_por_country <- table1 |>
  group_by(country) |>
  summarise(cases = sum(cases))
```

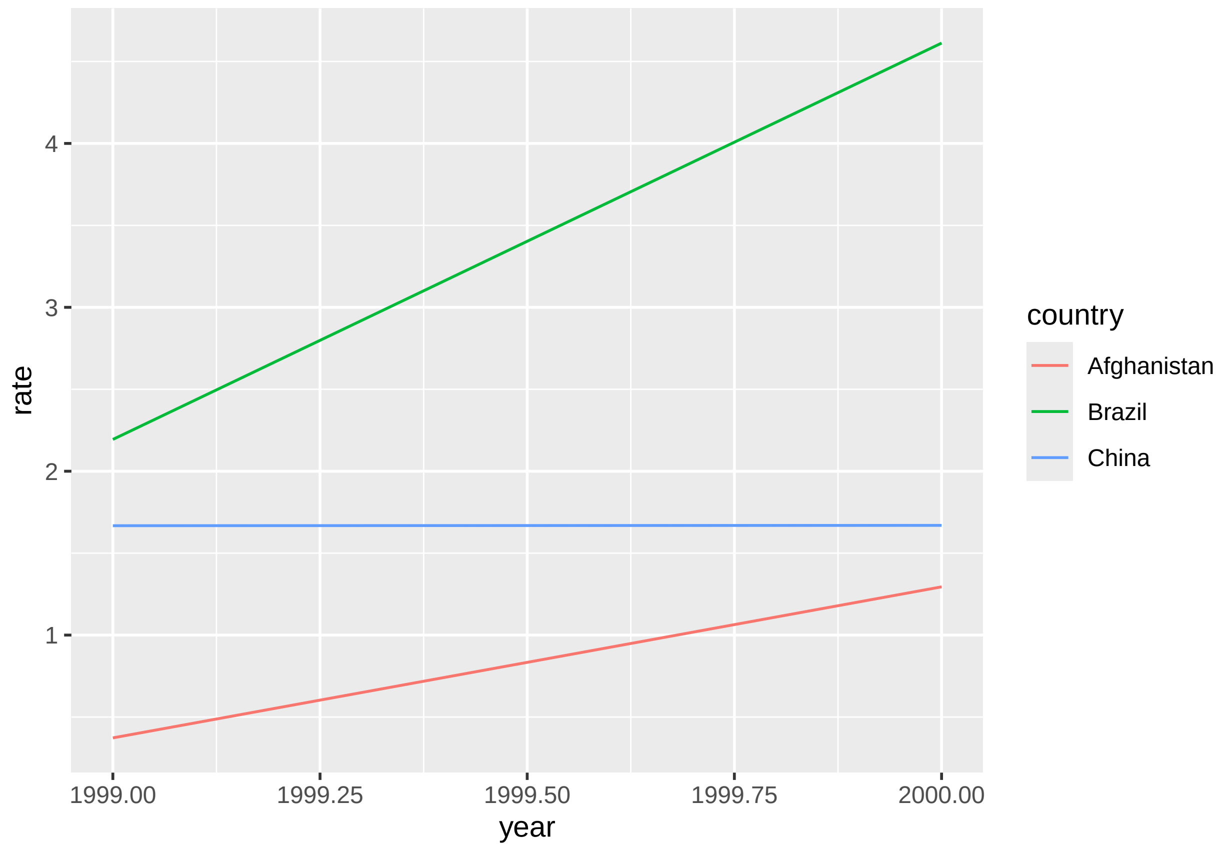
Exercicio 6

```
table1 |>
  select(country, year, cases) |>
  pivot_wider(
    names_from = year,
    values_from = cases
  )
```

```
## # A tibble: 3 x 3
##   country     ‘1999‘ ‘2000‘
##   <chr>        <dbl>  <dbl>
## 1 Afghanistan    745   2666
## 2 Brazil       37737  80488
## 3 China       212258 213766
```

Exercicio 7

```r
ggplot(taxas, aes(x = year, y = rate, color = country)) +
  geom_line()
```



Exercicio 8

```r
casos <- table2 |>
  filter(type == "cases") |>
  select(country, year, cases = count)

populacao <- table2 |>
  filter(type == "population") |>
  select(country, year, population = count)

taxas2 <- casos |>
  left_join(populacao, by = c("country", "year")) |>
  mutate(rate = cases / population * 10000)

taxas2
```

```
## # A tibble: 6 x 5
```

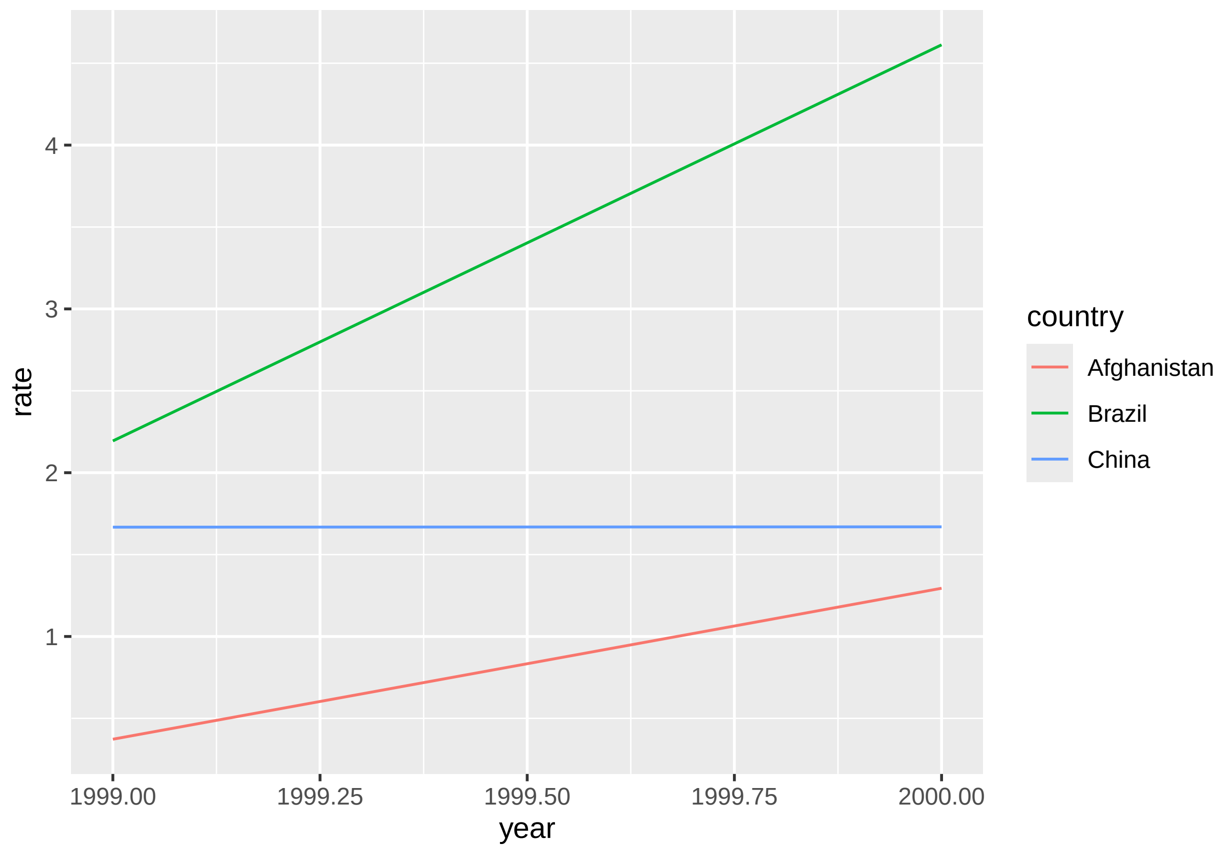
```
##   country      year  cases population  rate
##   <chr>       <dbl>  <dbl>      <dbl> <dbl>
## 1 Afghanistan  1999    745   19987071 0.373
## 2 Afghanistan  2000   2666   20595360 1.29
## 3 Brazil       1999  37737  172006362 2.19
## 4 Brazil       2000  80488  174504898 4.61
## 5 China        1999 212258 1272915272 1.67
## 6 China        2000 213766 1280428583 1.67
```

```
### Taxa utilizando `table4a` e `table4b`

casos4 <- table4a |>
  pivot_longer(
    cols = c(`1999`, `2000`),
    names_to = "year",
    values_to = "cases"
  )

populacao4 <- table4b |>
  pivot_longer(
    cols = c(`1999`, `2000`),
    names_to = "year",
    values_to = "population"
  )

taxas4 <- casos4 |>
  left_join(populacao4, by = c("country", "year")) |>
  mutate(rate = cases / population * 10000)

taxas4
```

```
## # A tibble: 6 x 5
##   country     year   cases population  rate
##   <chr>       <chr>  <dbl>      <dbl> <dbl>
## 1 Afghanistan 1999     745   19987071 0.373
## 2 Afghanistan 2000    2666   20595360 1.29
## 3 Brazil      1999   37737  172006362 2.19
## 4 Brazil      2000   80488  174504898 4.61
## 5 China       1999  212258 1272915272 1.67
## 6 China       2000  213766 1280428583 1.67
```

Exercicio 9

```
ggplot(taxas2, aes(x = year, y = rate, color = country)) +
  geom_line()
```



Exercicio 10

```
tidy4a <- table4a |>
  pivot_longer(
    cols = c(`1999`, `2000`),
    names_to = "year",
    values_to = "cases"
  )

tidy4a
```

```
## # A tibble: 6 x 3
##   country     year   cases
##   <chr>       <chr>  <dbl>
## 1 Afghanistan 1999     745
## 2 Afghanistan 2000    2666
## 3 Brazil      1999   37737
## 4 Brazil      2000   80488
## 5 China       1999  212258
## 6 China       2000  213766
```

Exercicio 11

```r
tidy4b <- table4b |>
  pivot_longer(
    cols = c(`1999`, `2000`),
    names_to = "year",
    values_to = "population"
  )

tidy4b
```

```
## # A tibble: 6 x 3
##   country     year  population
##   <chr>       <chr>      <dbl>
## 1 Afghanistan 1999    19987071
## 2 Afghanistan 2000    20595360
## 3 Brazil      1999   172006362
## 4 Brazil      2000   174504898
## 5 China       1999  1272915272
## 6 China       2000  1280428583
```

Exercicio 12

```r
tidy4 <- tidy4a |>
  left_join(tidy4b, by = c("country", "year"))

tidy4
```

```
## # A tibble: 6 x 4
##   country     year   cases population
##   <chr>       <chr>  <dbl>      <dbl>
## 1 Afghanistan 1999     745   19987071
## 2 Afghanistan 2000    2666   20595360
## 3 Brazil      1999   37737  172006362
## 4 Brazil      2000   80488  174504898
## 5 China       1999  212258 1272915272
## 6 China       2000  213766 1280428583
```

```r
# O comando em questao, left_join, combina duas ou mais tabelas utilizando uma
    #ou mais variaveis em comum como chave.
```

Exercicio 13

```r
tidy2 <- table2 |>
  pivot_wider(
    names_from = type,
    values_from = count
```

```r
  )

tidy2
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <dbl>      <dbl>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```

Exercicio 14

```r
tidy3 <- table3 |>
  separate(
    rate,
    into = c("cases", "population"),
    sep = "/",
    convert = TRUE
  )

tidy3
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <int>      <int>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```